MIL-C-46920B (AR)
24 August 1981
SUPERSEDING
MIL-C-46920A (MU)
31 January 1964

MILITARY SPECIFICATION

CHARGE, EXPELLING, FOR CARTRIDGE ILLUMINATING, 4.2 INCH, M335A2

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the loading, assembling and packing of one type of expelling charge for use with the Cartridge, 4.2 Inch Illuminating, M335A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this Specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGAGING DRAWINGS

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
8886594 - Expelling Charge Assembly

INSPECTION EQUIPMENT DRAWINGS

9221798 - Envelope Drawing for Leakage Testing

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 0-190 - Code of Federal Regulations Title 49 (Revised as of January 1, 1968)

The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1968 Edition and revisions) available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite "49 CFR 0-190 (Revised as of 1 January 1968).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 8886594 and associated drawings and with all requirements specified in applicable specifications and standards.

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3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. The components and assemblies shall be free from cracks, air bubbles, holes, blisters, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classificaion of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part	Drawing	Quantity
Cover, Expelling Charge (if applicable)	8886592	5*
Cover, Expelling Charge (if applicable)	9219239	5*
Cover, Expelling Charge (if applicable)	9219984	5*
Cup, Expelling Charge	8886593	5*
Expelling Charge Assembly	8886594	15

* Per each mold cavity

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46920B (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Expelling Charge Assembly and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cover, Expelling Charge (dwg. 8886592, 9219239, 9219984) (if applicable)				
	Examination for defects	5*		3.2	4.4.2.2
	Cup, Expelling Charge (dwg. 8886593)				
	Examination for defects	5*		3.2	4.4.2.1
	Expelling Charge Assembly (dwg. 8886594)				
	Examination for defects	15		3.2	4.4.2.3
	Moisture content of black powder	1		3.2	4.4.2.3/4.5.1
	Quick leak test	14		3.2	4.4.2.3/4.5.2

NOTES: *Per mold cavity

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46920B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	8886593
			NEXT HIGHER ASSEMBLY 8886594

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Inside diameter at open end		0.65%	3.2	Gage
202	Wall thickness		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Height of lip		0.65%	3.2	Gage
205	Lip thickness		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46920B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cover, Expelling Charge	1 OF 1	8886592,9219239 or 9219983
			NEXT HIGHER ASSEMBLY: 886594

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Thickness of flange (if applicable)		0.65%	3.2	Gage
204	Flange diameter (if applicable)		0.65%	3.2	Gage
205	Height (if applicable)		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46920B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8886594
4.4.2.3	Expelling Charge Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture content of black powder		*	3.2	4.5.1
102	Quick leak test		0.25%	3.2	4.5.2
Minor					
201	Marking misleading or unidentifiable including color code		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES: *The moisture content shall be determined at the time of loading. At least one sample shall be tested from each eight hours of production. If a sample fails to comply, the hours of production represented by the sample shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46920B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER (See 5.1 and 5.2)
4.4.2.4	Containers				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Container damaged so that contents are exposed or liable to be exposed		0.40%	5.1	Visual
Minor 201	Marking misleading or unidentifiable		0.65%	5.2	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Moisture content of black powder. This test shall be performed as specified in MIL-P-223.

4.5.2 Quick leak test. This test shall be perfomed as specified on dwg. 9221798. However, the leakage rate shall be no greater than 0.4 atmospheric cc/sec. in lieu of 0.01 atmospheric cc/sec. as cited on the drawing.

5. PACKAGING

5.1 Packaging and packing requirements

5.1.1 Level C. The expelling charges shall be packaged and packed in accordance with Code of Federal Regulations 49 CFR 0-190.

5.2 Marking

5.2.1 Level C. The marking shall be in accordance with Code of Federal Regulations 49 CFR 0-190 and dwg. 8796522. The DOT marking shall be "Black Powder".

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use in the 4.2 Inch, Illum. M335A2 and CS Smoke, XM630.

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6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approval. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is not under the cognizance of ARRADCOM.

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